

NS-WSG Boards - Parts Shopping List

Quantities cover 3 card builds + 1 riser set, with spares. BOM detail: hardware/ORDER.md

Boards ordered from PCBWay 2026-08-31 (10x riser, 5x card). (c) 2026 David Troy, in memory of Stephen Troy.

Qty	Part	Package / detail	Refs	Source / note
SEMICONDUCTORS & MODULES				
☐	3	Raspberry Pi Pico WH	RP2040 + WiFi, headers pre-soldered	module official reseller; ~\$7 ea
☐	6	SN74LVC245AN	DIP-20, octal bus xcvr	U1, U2 TI; DigiKey/Mouser
☐	6	SN74HCT541N	DIP-20, octal buffer	U3, U4 TI; DigiKey/Mouser
☐	3	SN74HCT138N	DIP-16, 3-to-8 decoder	U5 TI; DigiKey/Mouser
☐	3	PT8211-S	SOP-8, 16-bit stereo DAC	U6B LCSC/Alibaba; solders to the SOP land
SOCKETS & HEADERS				
☐	12	DIP-20 socket	0.3 in	U1-U4
☐	3	DIP-16 socket	0.3 in	U5
☐	3	DIP-8 socket	0.3 in	U6 alt only if using DIP PT8211/adaptor
☐	6	Female header 1x20, 0.1 in	Pico socket	module or cut from longer strips
☐	1	Male header strip 0.1 in	40-pin breakaway, x5 strips	J2 + risers risers need 2x15 each
PASSIVES (per-card refs; qty covers 3 builds + spares)				
☐	25	0.1 uF ceramic	50V, 0.1 in lead spacing	C1-C6, C10
☐	10	10 uF electrolytic	16V+ radial	C7, C8, C13
☐	10	47 uF electrolytic	16V+ radial	C9, C11, C12 audio coupling + VDD bulk
☐	5	10 ohm 1/4W	axial	R6
☐	5	1 k 1/4W	axial	R4
☐	5	10 k 1/4W	axial	R5
CONNECTORS & MISC				
☐	3	CUI SJ1-3523N	3.5 mm TRS jack, THT horizontal	J1 DigiKey/Mouser (CUI Devices)
☐	1	3.5 mm stereo cable or powered speakers		test audio
☐	1	Hookup wire 24-26 AWG	small spool	J2 mix-in
OPTIONAL / NICE TO HAVE				
☐	2	PAM8302 amp module	mono class-D breakout	drives a bare 8-ohm speaker from J1
☐	2	SOP-8 to DIP-8 adapter		U6 alt only if SOP-in-socket preferred
☐	1	Micro-USB cable	data-capable	Pico flashing if none on hand

Everything is through-hole except the PT8211-S (SOP-8, 1.27 mm pitch - easy drag solder).

TI DIP logic: verify 'N' suffix (PDIP) when ordering. Jack: CUI Devices SJ1-3523N exactly - the footprint matches its datasheet.